PARAMDEEP SINGH

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CAREER OBJECTIVE

Motivated B.Sc. Biotechnology student (SGPA 9.47; PSEB Merit List, top 0.1%) with a balanced skill set across the wet lab and the dry lab. Hands-on exposure to botanical extraction, fermentation, and PCR principles, backed by Python-based bioinformatics, BLAST-driven comparative genomics, and sequence analysis. Seeking a life science internship in molecular biology, microbiology, genomics, bioinformatics, or biotechnology R&D, where I can contribute analytical rigour and a strong work ethic while building advanced laboratory and computational expertise.

EDUCATION

B.Sc. Biotechnology | SGPA (Semester 1): 9.47 *2025 – 2028*
Chandigarh Group of Colleges (CGC University), Jhanjeri, Mohali | I.K. Gujral Punjab Technical University, Jalandhar

Senior Secondary (Class 12), Science: Physics, Chemistry, Biology | 97.4% *March 2025*
Punjab School Education Board (PSEB) | Rank 157 in PSEB Merit List

TECHNICAL SKILLS

Laboratory & Molecular Biology: PCR principles and DNA amplification, aqueous and botanical extraction, fermentation technology, microbiology, CRISPR fundamentals

Bioinformatics & Computational Biology: Biopython, BLAST, pairwise sequence alignment, comparative genomics, GenBank and FASTA handling, restriction enzyme analysis, virtual PCR optimization

Programming & Data: Python, Streamlit, SQL (basic), SQLite, HTML, CSS, data structures, biostatistics, Git/GitHub, VS Code

Research & Communication: Scientific writing, academic writing, research methodology, scientific presentation, event coordination, MS Word, MS PowerPoint

PROJECTS

Comparative Genomics of Pathogenic E. coli Strains *2026*
In-silico Computational Biology Project | Python, Biopython, BLAST

- Compared the genomes of E. coli O157:H7 Sakai and E. coli 55989 to identify genomic differences linked to pathogenicity.
- Performed whole-genome BLAST, genome-wide dot-plot analysis, circular genome comparison, and GenBank feature and annotation analysis using Biopython.
- Mapped the presence or absence of virulence-associated factors and examined the genomic context of Shiga-toxin-associated regions, revealing strain-specific differences in virulence gene content.

BioDNA-Inspector *2026*
Personal Bioinformatics Project | Python, Biopython, Streamlit

- Built a web-based DNA analysis application covering GC content, nucleotide composition, reverse complement, transcription, translation, and mutation generation.
- Integrated FASTA file upload, sequence comparison, pairwise alignment, and restriction enzyme analysis in a single interface.
- Added graphical visualisations and automated report generation to make sequence data easier to interpret for research and teaching.

BioNeemix: Plant-Based Antifungal Spray *2026*
Team Project, Department of Biotechnology, CGC University Mohali

- Formulated a plant-based antifungal spray combining Neem (azadirachtin) and Clove (eugenol); the combined action showed markedly higher antifungal efficacy than either extract alone.
- Carried out aqueous extraction by boiling, clove-water preparation, and colour extraction to isolate the active botanical compounds.
- Presented the work at Biotech Blooms Mini Project Day, demonstrating plant-derived compounds as an approach to fungal control.

VITALOS: Hospital Management System *2026*
Personal Software Project | Python, Streamlit, SQLite, ReportLab

- Developed a multi-module hospital management system with secure Email OTP login, patient registration, doctor management, appointment scheduling, and billing with PDF invoice generation.
- Built an interactive dashboard summarising patient, doctor, appointment, and billing data, with SQLite for storage and retrieval.

INTERNSHIPS & TRAINING

Online Certified Internship: Polymerase Chain Reaction (PCR) *July 2026*

Mathi LifeSciences | One-week programme

- Studied the principles and applications of PCR, including DNA amplification and related molecular biology techniques used in research and diagnostics.
- Consolidated molecular biology concepts through structured lectures and guided learning activities, complementing my in-silico PCR work.

CERTIFICATIONS & WORKSHOPS

Vaccine Designing & Development using AI (Workshop) *5 – 6 Oct 2026*

Amplikon Biosystems | Certificate of Participation

- Two-day workshop on practical knowledge and research skills in AI-assisted vaccine design and development.

NIeCer 103: Scientific Writing in Health Research | Score: 99% *August 2026*

ICMR-National Institute of Epidemiology, SWAYAM

Python for Everybody Specialization *June 2026*

University of Michigan (Coursera) | 5-course specialization

Introduction to Academic Research Specialization *June 2026*

University of California, Davis (Coursera) | 4-course specialization

Industrial Biotechnology *June 2026*

University of Manchester (Coursera)

LEADERSHIP & POSITIONS OF RESPONSIBILITY

Joint Secretary, VitaCore Club, CGC University *Sept 2026 – Present*

- Plan, coordinate, and execute multi-tier departmental and university events, handling logistics and student outreach.
- Liaise between faculty coordinators and student members to keep communication and event management running smoothly.
- Received a Certificate of Appreciation for coordinating the Microbe-to-Market: Microbial Product Development Challenge (24 Sept 2026).

Public Relations Officer, VitaCore Club, CGC University *2026 – Sept 2026*

- Promoted club events and activities through outreach and communication, growing student engagement and participation.
- Supported event coordination and publicity; received a Certificate of Appreciation for coordinating Campus Ferment Cafe (11 Sept 2026).

HONORS & AWARDS

3rd Position, Campus Ferment Cafe: Where Science Meets Taste *11 Sept 2026*

Department of Biotechnology, CGC University, Mohali

- Co-developed a fermented beetroot barfi through a three-day fermentation process, gaining practical exposure to fermentation-based food processing and product development.

PSEB Merit Certificate: Top 0.1% of Students *May 2025*

- Recognised for securing a place in the top 0.1% of students statewide.

ACTIVITIES & SCIENTIFIC EXPOSURE

Microbe-to-Market: Microbial Product Development Challenge *24 Sept 2026*

Presented the project MicroMend in a microbial product development challenge organised by VitaCore Club, CGC University.

National Conference: Fungi for Society and Sustainability (FSS-2026) *10 Oct 2026*

Attended the World Fungus Day conference hosted by Dr. C. V. Raman University, gaining exposure to current work on fungi and sustainable development.

Educational Visit: CSIR-IMTECH, Chandigarh *2026*

Observed a professional biomedical research environment and the role of advanced facilities in life-science research.

Biotech Blooms: Mini Project Day, CGC University *2026*

Presented BioNeemix and practised scientific presentation and project demonstration.

Fermentation Technology Workshop, CGC University *2026*

Took part in a workshop on fermentation technology and its industrial applications.

Ideathon: Biotechnology Innovation, CGC University *2026*

Developed problem-solving and pitching skills while exploring biotechnology-driven solutions.

Scientific Session: A Conversation About Curiosity, Discovery and a Life in Science *2026*

Explored perspectives on careers in the biological and medical sciences at Open Lab.